

WONG WEI MING

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EDUCATION

Nanyang Technological University, Singapore

Aug 2023 - Present

Bachelor of Engineering (Honours), Computer Engineering & Data Analytics (Minor in Entrepreneurship)

Hong Kong University of Science and Technology, Hong Kong

Jan 2026 - Present

Semester Exchange — B.E. Computer Engineering

- Coursework: Machine Learning, Deep Learning in Computer Vision, Intelligent Robot & Embodied AI, Embedded AI Systems, Intro to Aerial Robotics

ACHIEVEMENTS

- NTU President Research Scholar** (URECA) — Undegraduate Research
- Semifinalist** (Top 10) & **Best New All-Rounders** — Robonation Robosub 2025 (International Level)
- Champion** — Singapore AUV Challenge 2025 (International Level)
- Finalist** — HacX @ HTX 2025
- Finalist** — NTU Escendo @ Garage EEE 2025 (School Level)
- Rank 9** — Ocean Exploration Video Challenge, MATEROV 2024 (International Level)

SKILLS

- Programming Languages:** Python, C, C++, Java
- Robotics & Autonomous Systems:** ROS1/ROS2, Nav2, Isaac ROS, NVBlox, ArduPilot, Behavior Tree, SLAM, PID Control
- AI & Computer Vision:** PyTorch, YOLO, OpenCV, Roboflow, TensorRT, ONNX
- Simulation:** Unity, Isaac Sim, Gazebo
- Tools:** Git, Docker, Linux/Ubuntu, WSL
- Hardware:** NVIDIA Jetson, Raspberry Pi, Pixhawk, STM32, ESP32, Intel RealSense, Soldering, Electronic Assembly, System Integration

EXPERIENCE

Mecatron (AUV Team in NTU)

Aug 2024 - Present

Perception Team Lead

- Led a 10-member perception software team in a 60-member club for competitions using ROS2 and computer vision.
- Developed software stack with Behavior Tree for autonomous task sequencing in Robosub and SAUVC; performed SITL testing in Unity.
- Performed electronic assembly and system integration for the AUV hardware stack.
- Conducted 5+ robotics workshops (computer vision, ROS, robotic arm, eduboot) for 30-40 NTU students per session.

SkyrisAI

Dec 2025 - Jan 2026

ROS Developer (Details limited due to NDA)

- Contributed to the development of a novel autonomous platform, focusing on high-level system architecture and modular design.
- Designed the ROS 2 framework and simulation environments to enable seamless interaction between AI models and robotic hardware.
- Researched SLAM pipelines and AI integration strategies to enhance the robot's environmental understanding.

Home Team Science & Technology (HTX)

May 2025 – Dec 2025

Artificial Intelligence & Robotics Engineer Intern

- Integrated Isaac ROS with Nav2 and Intel RealSense D435i on Jetson AGX Orin for autonomous visual navigation.
- Implemented real-time 3D reconstruction and SLAM using NVBlox from Isaac ROS with Nav2 on AGX Orin.
- Integrated ROSA (ROS Agent) to enable LLM-driven sequential planning and ROS tool calling for robot control.
- Developed a unified core stack based on Jetson and Pixhawk (ArduPilot), tested on 2 UGV platforms and designed for cross-platform deployment on AUV, UAV, and UGV systems.

Machine Learning and Data Analytics Club (MLDA)

Oct 2024 – May 2025

Robotics Software Developer

- Developed ROS1 Noetic-based control for DJI Tello Edu Drone in a 4-member team.
- Implemented human detection using YOLOv8, achieving ~80% detection accuracy.

PROJECTS & RESEARCH

Undergraduate Research Programme (URECA), CCDS, NTU

Aug 2024 – May 2025

Undergraduate Research

- Deployed and benchmarked VILA 1.5-3B Vision-Language Model on Jetson Orin NX (16GB) for real-time underwater situational awareness on AUVs.

- Integrated VLM inference with the ROS system to achieve environmental understanding and action planning for autonomous underwater navigation.
- Applied 4-bit quantization via NVIDIA Jetson Container, achieving ~3.1s inference latency at 25W power mode for edge deployment.

Ocean Exploration Video Challenge (Rank 9), MATEROV 2024

July 2024 - Aug 2024

Software Team (Mechatronics)

- Developed YOLOv10-based brittle star detection software for dark, murky underwater environments.
- Prepared datasets by combining real-world images with synthetic data generated in Unity, improving detection accuracy by ~30%.
- Achieved 80-90% accuracy using a moving slicer inference method, balancing speed and precision.
- Leveraged Python, OpenCV, and Roboflow for data augmentation and object detection model training.

VOLUNTEERING

CCDS Tech for Good Hackathon 2026 — Sub-committee Member

Oct 2025 - Feb 2026

- Prepared LangChain workshop content and training materials for ~40 hackathon participants.

NTU CCDS Student Ambassador

Aug 2024

- Conducted programming workshops for BIM Scholar Programme students.

CO-CURRICULAR ACTIVITIES

NTU CCDS, COE & NTUpreneur – Student Ambassador

Aug 2024 - Present

NTU College of Computing and Data Science (CCDS) Orientation – Chief Programmer

Aug 2024 - Present

NTU Sports Club - Programmer

Aug 2023 - Aug 2024

NTU Taekwondo Varsity Team

Aug 2023 - May 2024